

ENGN 1931Y: Control Systems Engineering

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0.1 Sept 10

Control system: a mechanism that alters the future state of a system

Control theory: a strategy to achieve the desired system performance

- Alternatively, the study of how to regulate and manage the behavior of dynamic system using feedback control loop

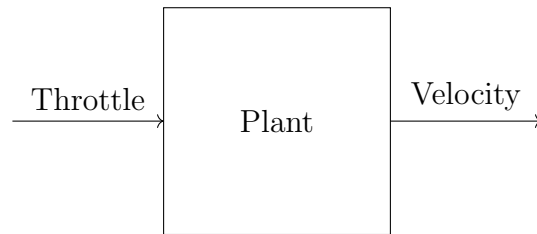
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Types of Control Systems

- Logic control (e.g. programmable logic controller)
- On-off control (e.g. light switch)
- Feedback control

In this class we will focus on feedback control systems and represent almost everything with block diagrams abstracting the system into inputs and outputs.

Example: A car block diagram

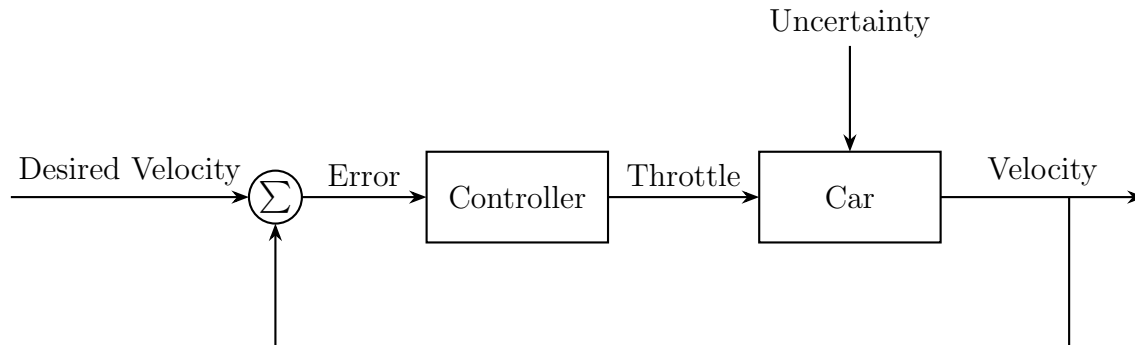


Open loop control: system without feedback.

- In the car example, we can imagine a cruise control system that sets the throttle to a fixed value. Without incorporating knowledge of dynamics or feedback, we cannot compensate for uncertain masses, varying environmental controls, or imprecise models

Closed loop control: system with feedback used to control despite our incomplete knowledge (though this trades off stability, response speed, etc.)

- In the car example, we update the throttle command based on a measurement of the current speed



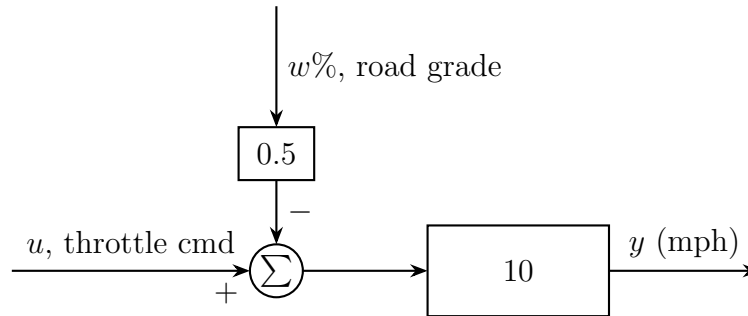
Any block diagram must involve:

- Desired reference
- Controller
- Plant

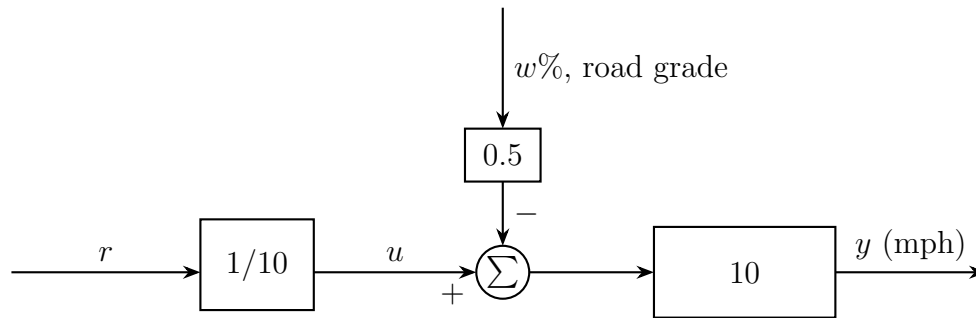
- Sensor
- Feedback
- Uncertainty

Example: Consider the system in a car where 1 degree throttle change maps to 10 mph change in speed and a 1% change in road grade maps to -5 mph change in speed. How do we relate road grade to throttle change? (Trivially, 1% change equals 0.5 degree throttle)

How might we draw this controller?



The naive open loop controller ignores the disturbance and simply maps the desired speed to 1/10 of that value in throttle so $r = y$ if $w = 0$.



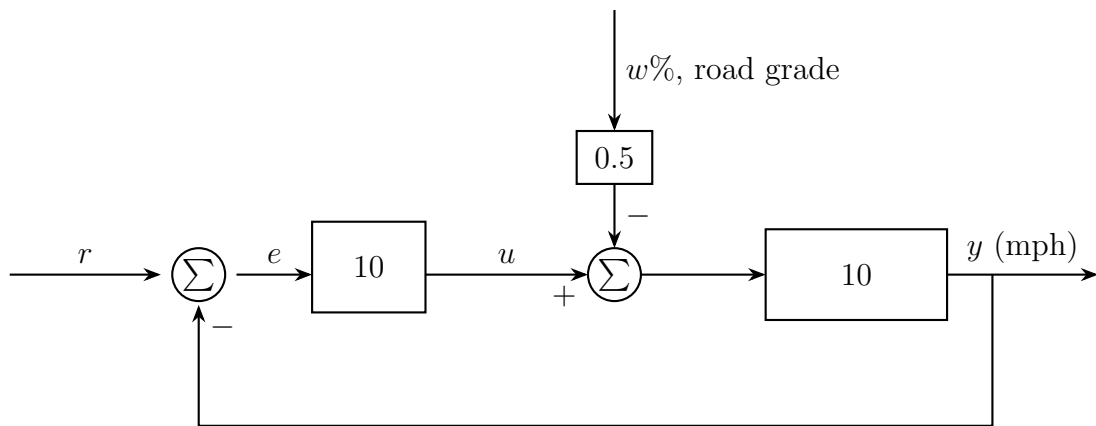
Hence the actual control exerted is

$$\begin{aligned}
 y &= r - 0.5(10)w \\
 &= r - 5w \\
 e &= r - y \\
 &= r - (r - 5w) \\
 &= 5w
 \end{aligned}$$

so our % speed error is $500w/r$. At $r = 65$ mph and $w = 1\%$, we have a 7.7% error in speed.

Remark: in this example we had to choose 1/10 as the gain of the controller to achieve $r = y$ when $w = 0$. This required knowledge of the plant dynamics. In general, we may not know the plant dynamics and hence cannot choose a controller gain to achieve this. This motivates the use of feedback control.

Let's try this again with feedback control.



If we arbitrarily set the gain at 10, our system is governed by

$$\begin{cases} y = 10(u - 0.5w) \\ u = 10(r - y) \end{cases} \implies y = 100r - 100y - 5w \implies y = \frac{100}{101}r - \frac{5}{101}w \approx 0.99r - 0.05w$$

If $w = 1$ and $r = 65$,

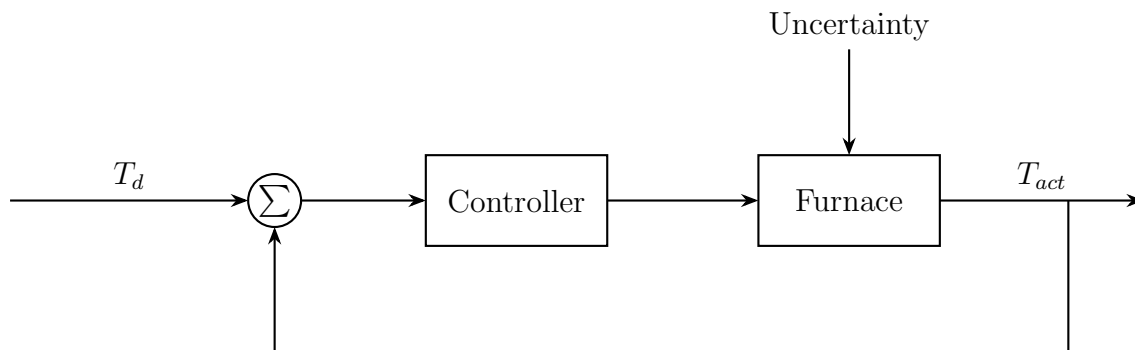
$$e = y - r = \frac{r}{101} + \frac{5}{101}w$$

and the error percentage is

$$e\% = \frac{100e}{r} = \frac{0.01r - 0.05}{r} \approx 0.076\%$$

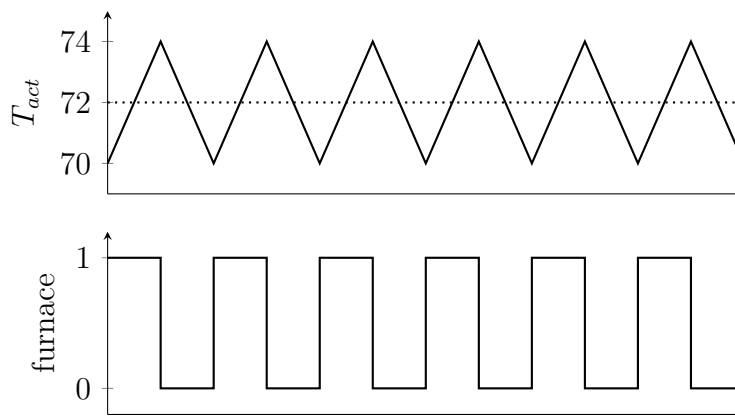
What if we set the gain to 1000? Wouldn't that send the error to 0.0007%? Not necessarily. This can lead to instability and oscillations.

Example: Consider a house with a heater and a thermostat.



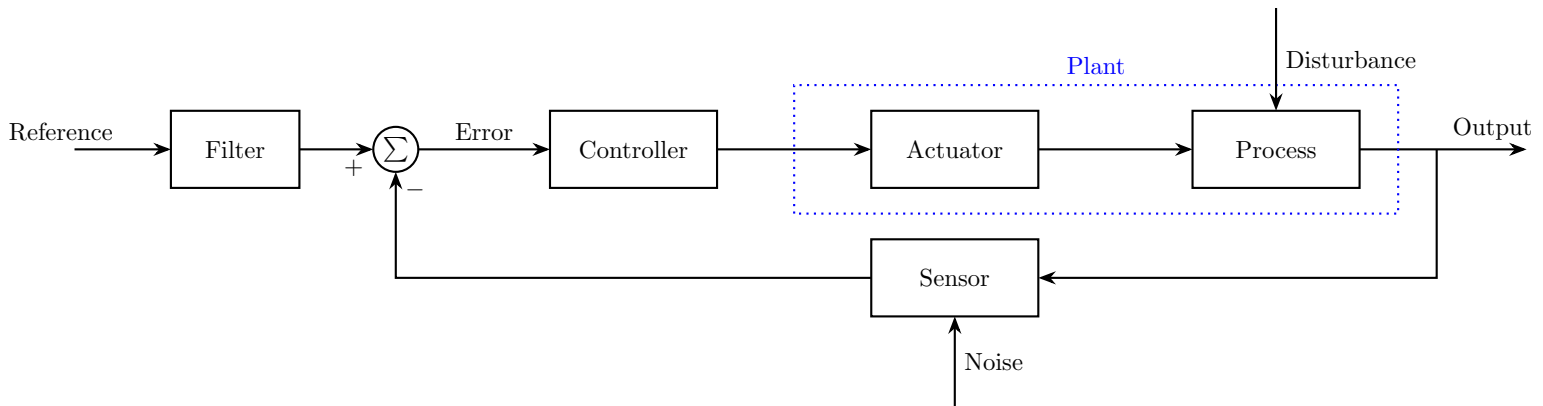
$$e = T_d - T_{act}$$

Notice here the furnace can only turn on and off so we have a delay in the system response. This results in the actual temperature sawtoothing around the setpoint:



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Recall: last time we introduced a general block diagram for a *Single-input single-output* (SISO) feedback control system.



Characteristics of a well-designed control system:

- *Stability*: the system should not diverge or oscillate uncontrollably
- *Tracking*: the system should follow the desired reference accurately
- *Disturbance rejection*: the system should be able to maintain performance in the presence of disturbances
- *Robustness*: the system should be able to handle model uncertainties and variations in system parameters

As we will see, each box in the block diagram can be represented by a mathematical model (in fact an ODE), and we can analyze the overall system behavior using these models.

Laplace Transform: for a bounded function $f(t)$, the Laplace transform is defined as

$$\mathcal{L}[f(t)] = F(s) = \int_0^{\infty} f(t) e^{-st} dt$$

where $s = \sigma + j\omega$ is a complex variable.

(*Remark:* we define the integral from 0 to ∞ because we are only interested in causal systems, i.e. systems that do not depend on future inputs.)

Examples:

$$\mathcal{L}[u(t)] = 1/s$$

($u(t)$ unit step function)

$$\mathcal{L}[tu(t)] = 1/s^2$$

$$\mathcal{L}[t^2u(t)] = 1/s^3$$

$$\mathcal{L}[\sin(\omega t)u(t)] = \frac{\omega}{s^2 + \omega^2}$$

$$\mathcal{L}[\cos(\omega t)u(t)] = \frac{s}{s^2 + \omega^2}$$

$$\mathcal{L}[\delta(t)] = 1$$

($\delta(t)$ Dirac delta function)

In Matlab, we can compute these transforms using the **Symbolic Math Toolbox** and

```
1 syms s t w
2 Laplace(cos(w * t))
```

Properties of the Laplace transform:

1. Linearity

$$\mathcal{L}[af(t) + bg(t)] = aF(s) + bG(s)$$

2. Frequency shift

$$\mathcal{L}[e^{-at}f(t)] = F(s + a)$$

3. Time shift

$$\mathcal{L}[f(t - T_d)] = F(s)e^{-T_d s}$$

4. Scaling

$$\mathcal{L}[f(at)] = \frac{1}{a}F\left(\frac{s}{a}\right)$$

5. Integration/differentiation

$$\begin{aligned}\mathcal{L}\left[\frac{d^n f(t)}{dt^n}\right] &= s^n F(s) + s^{n-1}f(0) + s^{n-2}f'(0) + \dots + f^{(n-1)}(0) \\ \mathcal{L}\left[\int_0^t f(t) dt\right] &= \frac{1}{s}F(s)\end{aligned}$$

6. Behavior under convolution

$$\mathcal{L}[f(t) \star g(t)] = \mathcal{L}\left[\int_0^t f(t - \tau)g(\tau) dt\right] = F(s)G(s)$$

7. Final/initial values

$$\lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} sF(s)$$

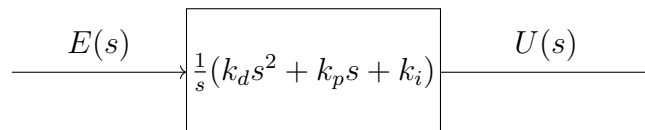
Proportional-Integral-Derivative Control: For $k_p, k_i, k_d \in \mathbb{R}$.

$$\begin{aligned}u(t) &= k_p e(t) + k_i \int_0^t e(\tau) d\tau + k_d \frac{de(t)}{dt} \\ U(s) &= k_p E(s) + k_i \frac{E(s)}{s} + k_d s E(s)\end{aligned}$$

Notice by grouping terms,

$$\frac{U(s)}{E(s)} = \frac{k_d s^2 + k_p s + k_i}{s}$$

so by using the Laplace transform, we can represent the PID controller as a polynomial transfer function



Exercise: Suppose there is a mass-on-spring system governed by

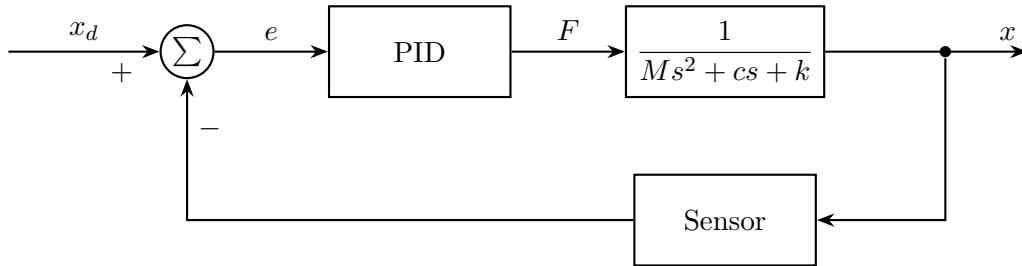
$$F = M\ddot{x} + c\dot{x} + kx$$

What is the transfer function?

$$\begin{aligned}\mathcal{L}[F] &= M(s^2X(s) - sx(0) - x'(0)) + c(sX(s) - x(0)) + kX(s) \\ &= (Ms^2 + cs + k)X(s)\end{aligned}$$

$$\frac{X(s)}{F(s)} = \frac{1}{Ms^2 + cs + k}$$

This will allow us to control the system by a simple block diagram



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Example: Use the Laplace transform to solve the following ODE:

$$\begin{cases} \ddot{y} + 5\dot{y} + 6y = 0 \\ y(0) = 2 \\ \dot{y} = 3 \end{cases}$$

$$\begin{aligned}\mathcal{L}[\ddot{y} + 5\dot{y} + 6y] &= \mathcal{L}[0] \\ s^2Y(0) - sy(0) - \dot{y}(0) + 5(sY(s) - y(0)) + 6Y(s) &= 0 \\ (s^2 + 5s + 6)Y(s) - 2s - 3 - 10 &= 0 \\ Y(s) &= \frac{2s + 13}{s^2 + 5s + 6} \\ y(t) &= \mathcal{L}^{-1}\left[\frac{2s + 13}{s^2 + 5s + 6}\right]\end{aligned}$$

In this example,

$$Y(s) = \frac{2s + 13}{s^2 + 5s + 6}$$

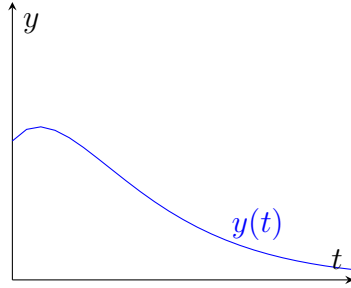
is our **transfer function**. We call the numerator our **zeroes** and the denominator our **poles**.

0.4.1 Inverse Laplace Transform

We use the **inverse laplace transform** to go from the transfer function back to the time domain.

In this problem, we need to use *partial fractions*:

$$\begin{aligned}\frac{2s+13}{(s+2)(s+3)} &= \frac{A}{s+2} + \frac{B}{s+3} \\ 2s+13 &= A(s+3) + B(s+2) \\ 2s+13 &= (A+B)s + (3A+2B) \\ \begin{cases} A+B &= 2 \\ 3A+2B &= 13 \end{cases} &\implies A=9, B=-7 \\ y(t) &= \mathcal{L}^{-1} \left[\frac{9}{s+2} - \frac{7}{s+3} \right] \\ &= \boxed{9e^{-2t} - 7e^{-3t}}\end{aligned}$$



Example:

$$F(s) = \frac{1}{s(s+0.1)^2} = \frac{K_1}{s} + \frac{K_2}{(s+0.1)^2} + \frac{K_3}{(s+0.1)}$$

Choosing K_1 and K_2 to cancel the denominators, we have

$$\begin{aligned}K_1 &= \frac{1}{(0.1)^2} = 100 \\ K_2 &= \frac{1}{s} \Big|_{s=-0.1} = -10\end{aligned}$$

and then for K_3 we just need to solve for the remaining constant. For ease of calculation, let $s = 1$:

$$\begin{aligned}\frac{1}{s(s+0.1)^2} \Big|_{s=1} &= \left[\frac{K_1}{s} + \frac{K_2}{(s+0.1)^2} + \frac{K_3}{(s+0.1)} \right]_{s=1} \\ \frac{1}{1(1.1)^2} &= 100 + \frac{-10}{(1.1)^2} + \frac{K_3}{1.1} \\ K_3 &= \frac{1}{1.1} - 100 + \frac{10}{(1.1)^2} \approx -90.74\end{aligned}$$

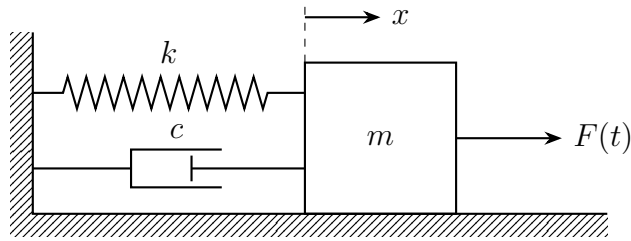
Note that when using partial fractions, we must have a term for each order of the repeated root.

This gives rise to a general transfer function of the form

$$\begin{aligned}H(s) &= \frac{a_n s^n + a_{n-1} s^{n-1} + \dots + a_1}{s^m + b_{m-1} s^{m-1} + \dots + b_1} \\ &= \frac{N(s)}{(s^2 + 2\zeta\omega_n s + \omega_n^2) \prod_{j=3}^n \frac{1}{s-s_j}}\end{aligned}$$

Example: With null initial conditions, the mass-spring-damper system is governed by

$$m\ddot{x} + c\dot{x} + kx = F(t)$$



The transfer function is

$$\frac{X(s)}{F(s)} = \frac{1}{ms^2 + cs + k}$$

We want to represent it as a **second order system** of the form

$$s^2 + 2\zeta\omega_n s + \omega_n^2$$

Notice

$$\frac{1}{ms^2 + cs + k} = \frac{1/m}{s^2 + \frac{c}{m}s + \frac{k}{m}}$$

Now $\omega_n = \sqrt{k/m}$ is the **natural frequency** while $2\zeta\omega_n = c/m$ is the **damping ratio**.

This second order form is essential because it allows us to decompose all higher order systems into a product of second order systems and first order systems.

Now consider the poles (the roots of the denominator $s^2 + 2\zeta\omega_n s + \omega_n^2$). What happens under different values of ζ ?

$$\begin{cases} \zeta = 0 & \implies s_1, s_2 = \pm j\omega_n \\ 0 < \zeta < 1 & \implies s_1, s_2 = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2} \\ \zeta = 1 & \implies s_1, s_2 = -\omega_n \\ \zeta > 1 & \implies s_1, s_2 \in \mathbb{R} : s_2 < \omega_n < s_1 \end{cases}$$

Exercise: Find the inverse Laplace transform of

$$\frac{5}{(s+1)(s^2+4s+8)}$$

First compare

$$s^2 + 4s + 8 = s^2 + 2\zeta\omega_n s + \omega_n^2$$

so $\omega_n = 2\sqrt{2}$ and $\zeta = 1/\sqrt{2}$.

From our analysis above, this means we have two complex conjugate poles at

$$s = -2 \pm j\sqrt{8 - \frac{8}{\sqrt{2}}}$$

Hence to get our inverse laplace, it suffices to write our function in the form

$$\begin{aligned} H(s) &= \frac{A}{s^2 + 2\zeta\omega_n s + \omega_n^2} \\ &= \frac{A}{s^2 + 2\zeta\omega_n s + \omega_n^2 + \zeta^2\omega_n^2 - \zeta^2\omega_n^2} \\ &= \frac{A}{(s^2 + \zeta\omega_n)^2 + \omega_n^2(1 - \zeta^2)} \end{aligned}$$

which has general inverse laplace transform

$$H(s) = \frac{A}{s^2 + 2\zeta\omega_n s + \omega_n^2} \iff h(t) = Ae^{-\zeta\omega_n t} \cos(\omega_n t \sqrt{1 - \zeta^2})$$

In our case,

$$\begin{aligned} H(s) &= \frac{5}{(s+1)(s^2+4s+8)} \\ &= \frac{k_1 s + k_2}{s^2 + 4s + 8} + \frac{k_3}{s+1} \end{aligned}$$

We can solve for k_1, k_2, k_3 by multiplying both sides by the denominator and choosing convenient values of s .

$$\begin{aligned} 5 &= (k_1 s + k_2)(s+1) + k_3(s^2 + 4s + 8) \\ s = -1 &\implies 5 = k_3(1 - 4 + 8) \\ &\implies k_3 = 5/5 = 1 \\ s = 0 &\implies 5 = k_2 + 8k_3 \\ &\implies k_2 = -3 \\ s = 1 &\implies 5 = 2(k_1 - 3) + 13 \\ &\implies k_1 = -1 \end{aligned}$$

All together,

$$\begin{aligned} F(s) &= \frac{1}{s+1} - \frac{s+3}{s^2+4s+8} \\ &= \frac{1}{s+1} - \frac{s+2}{(s+2)^2+4} - \frac{1}{2} \frac{2}{(s+2)^2+4} \end{aligned}$$

via completing the square.

From an inverse laplace table, we have

$$\begin{aligned} \mathcal{L}^{-1} \left[\frac{1}{s+1} \right] &= e^{-t} \\ \mathcal{L}^{-1} \left[\frac{s+2}{(s+2)^2+4} \right] &= e^{-2t} \cos(2t) \\ \mathcal{L}^{-1} \left[\frac{2}{(s+2)^2+4} \right] &= e^{-2t} \sin(2t) \end{aligned}$$

and

$$f(t) = e^{-t} + e^{-2t} [\cos(2t) + \sin(2t)]$$

This general method works if the denominator is of higher order than the numerator. If the numerator is of higher order, we can use polynomial long division to reduce the order of the numerator.

Example:

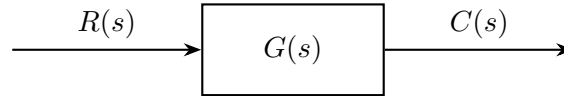
$$\frac{s^2 + 4s + 16}{s^2 + 5s + 25} = 1 - \frac{s + 9}{s^2 + 5s + 25}$$

but this is a problem! $\mathcal{L}^{-1}[1] = \delta(t)$ which means the system is returning an impulse from no input.

0.4.2 Block Diagrams

We have three primary ways to connect systems together: *series*, *parallel*, and *feedback*.

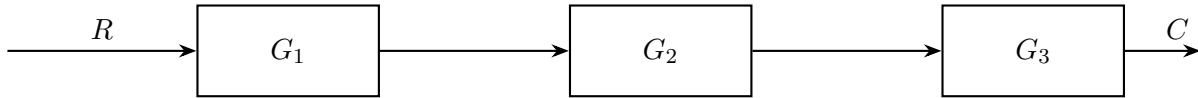
The simplest series block diagram is



which is equivalent to

$$C(s) = G(s) \cdot R(s)$$

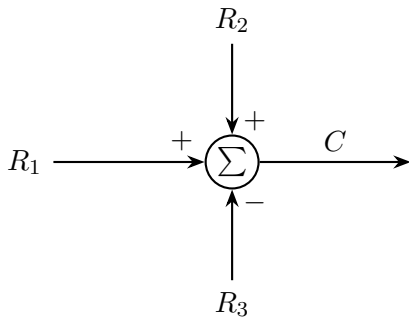
This could also look like a *cascade*:



$$C = G_1 \cdot G_2 \cdot G_3 \cdot R$$

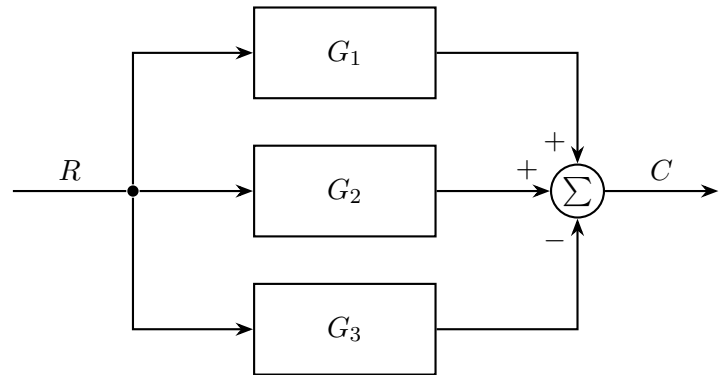
Parallel block diagrams appear in two forms:

Multi-input:



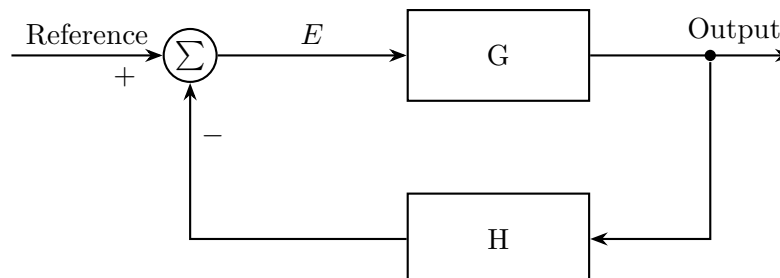
$$C = R_1 + R_2 - R_3$$

Multi-path:



$$C = (G_1 + G_2 - G_3)R$$

Feedback block diagrams are used to represent systems with feedback loops.

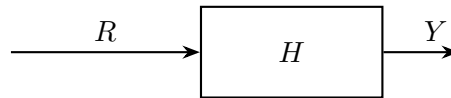


What is the transfer function of this system?

$$\begin{cases} E = R - CH \\ C = GE \end{cases} \implies C = G(R - CH) \implies C = \frac{GR}{1 + GH} \implies \frac{C}{R} = \frac{G}{1 + GH}$$

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Recall: For a block diagram



the transfer function is

$$H(s) = \frac{Y(s)}{R(s)}$$

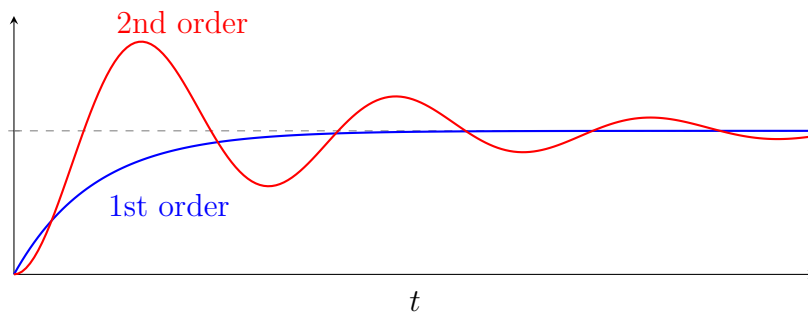
and the transfer function can be of any order:

$$H(s) = \frac{a_0 s^n + a_1 s^{n-1} + \dots + a_{n-1} s + a_n}{b_0 s^m + b_1 s^{m-1} + \dots + b_{m-1} s + b_m}$$

If $m > n$, we can use partial fractions to decompose the transfer function into a sum of first (a system with real poles) and second order systems (those with complex poles).

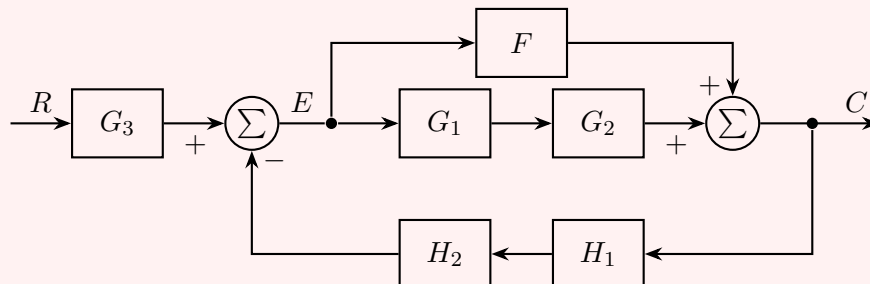
If $m = n$, the transfer function will be of the form $H(s) = 1 + \dots$

In general, a first order system will nicely converge to a steady state value, while a second order system will oscillate around the steady state value.



We can always represent the poles of a second order system in the form $s^2 + 2\zeta\omega_n s + \omega_n^2$ for $0 \leq \zeta \leq 1$. (If $\zeta = 1$, the system will have two real poles).

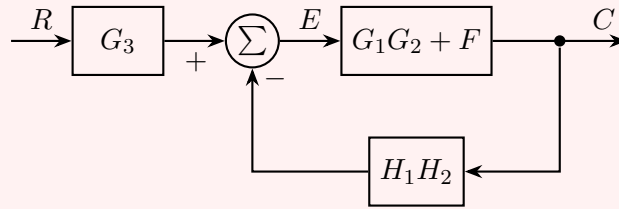
Exercise: What is the transfer function of the following system?



$$\begin{cases} E = RG_3 - H_2H_1C \\ C = EF + EG_1G_2 \\ \quad = E(F + G_1G_2) \end{cases}$$

$$\begin{aligned} C &= (RG_3 - H_2H_1C)(F + G_1G_2) \\ &= RG_3(F + G_1G_2) - H_2H_1C(F + G_1G_2) \\ C[1 + H_2H_1(F + G_1G_2)] &= RG_3(F + G_1G_2) \\ \frac{C}{R} &= \frac{G_3(F + G_1G_2)}{1 + H_2H_1(F + G_1G_2)} \end{aligned}$$

We can also use block diagram reduction to simplify the system. First, we can combine G_1 and G_2 in series, then combine H_1 and H_2 in series, then combine the two feedback loops into one, and finally combine G_3 with the rest of the system in series.

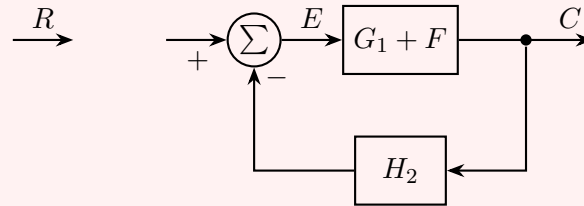


Which, using the general formula, gives

$$\frac{C}{G_3R} = \frac{F + G_1G_2}{1 + (H_1H_2)(F + G_1G_2)} \implies \frac{C}{R} = \frac{G_3(F + G_1G_2)}{1 + (H_1H_2)(F + G_1G_2)}$$

In the example above, we call the F branch a *feedforward* branch, while the H_1H_2 branch is a *feedback* branch. G_1G_2 is assumed to be the *plant control*.

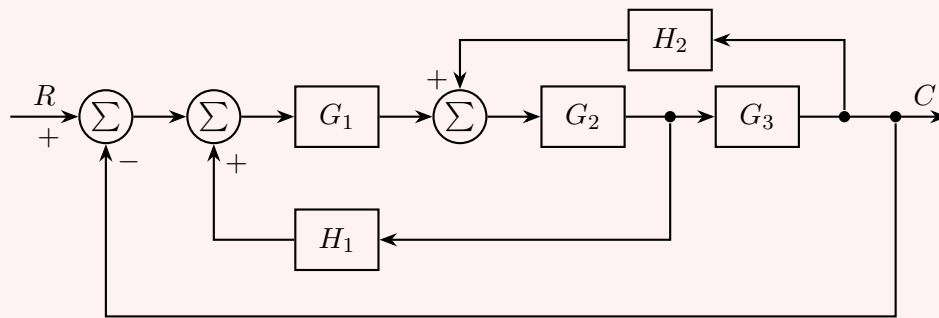
Exercise: Find the transfer function of the system



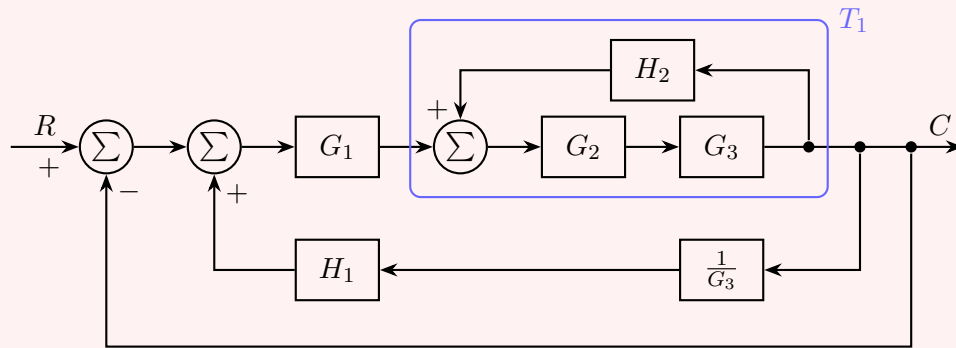
where $F = 1$, $G_1 = \frac{5}{s+5}$, $H_2 = \frac{7}{s+7}$,

$$\begin{aligned}
 \frac{C}{R} &= \frac{G_1 + F}{1 + H_2(G_1 + F)} \\
 &= \frac{\frac{5}{s+5} + 1}{1 + \frac{7}{s+7} \left(\frac{5}{s+5} + 1 \right)} \\
 &= \frac{\frac{5+s+5}{s+5}}{\frac{(s+7)(s+5) + 7(5+s+5)}{(s+7)(s+5)}} \\
 &= \frac{(s+7)(s+10)}{(s+7)(s+5) + 7(s+10)} \\
 &= \frac{(s+7)(s+10)}{s^2 + 19s + 105}
 \end{aligned}$$

Exercise: Find the transfer function of the system



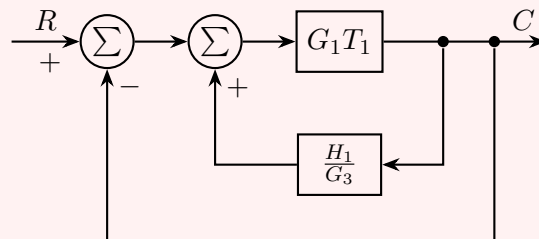
The trick is to simplify:



Inner block:

$$T_1 = \frac{G_2 G_3}{1 - G_2 G_3 H_2}$$

(notice the flipped sign in the denominator because the feedback is positive).

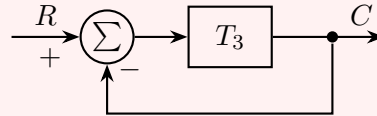


Then letting,

$$T_2 = \frac{H_1}{G_3}$$

we have

$$T_3 = \frac{G_1 T_1}{1 - G_1 T_1 T_2}$$



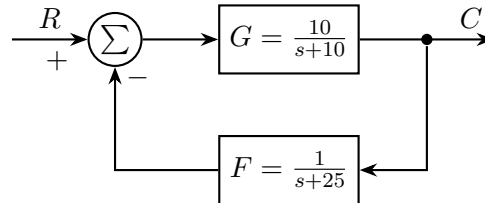
And finally,

$$\frac{C}{R} = \frac{T_3}{1 + T_3}$$

As you can see, block diagram reduction is a very useful tool for simplifying complex systems. It allows us to break down the system into smaller, more manageable parts, and then combine them to find the overall transfer function. However, this is still a very tedious process, and it is easy to make mistakes.

Let's use Matlab instead.

Example:

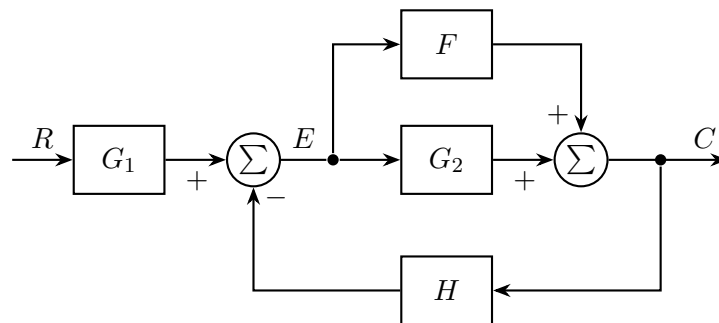


```
1 s = tf('s');  
2 G = 10/(s + 10);  
3  
4 step(G) % gets the step response of the system  
5 impulse(G) % gets the impulse response of the system  
6  
7 F = 1/(s + 25);  
8 CLTF = feedback(G, F) % gets the closed loop transfer function of the system
```

This will output

$$CLTF = \frac{10s + 250}{s^2 + 35s + 250}$$

Example:



We can define $Q_1 = F + G_2$ so the system can be solved with

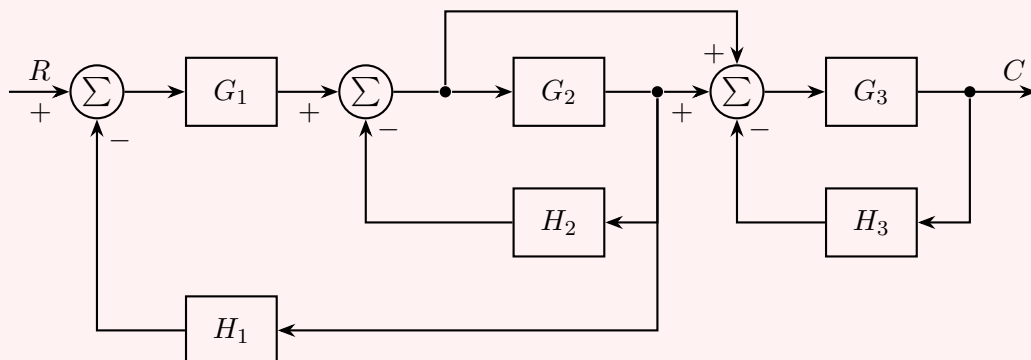
```
1 s = tf('s');  
2 syms F G1 G2 H;  
3 Q = G + G2;  
4
```

```

5 M = feedback(Q, H);
6 CLTF = Series(M, G1)
7
8 ltview(CLTF) % opens the linear system analyzer

```

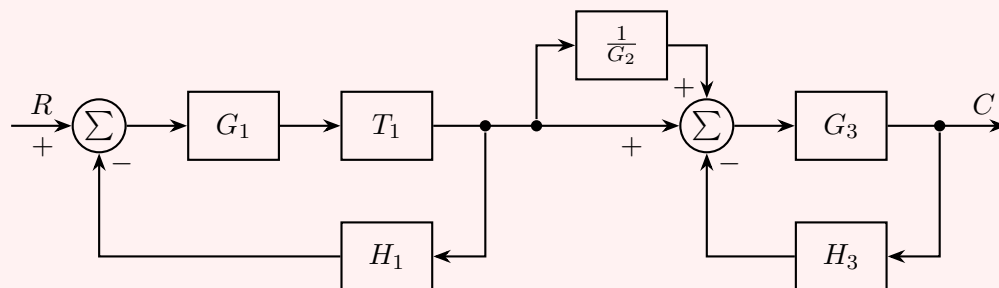
Exercise: Find $\frac{C}{R}$ for the system



Letting

$$T_1 = \frac{G_2}{1 + G_2 H_2}$$

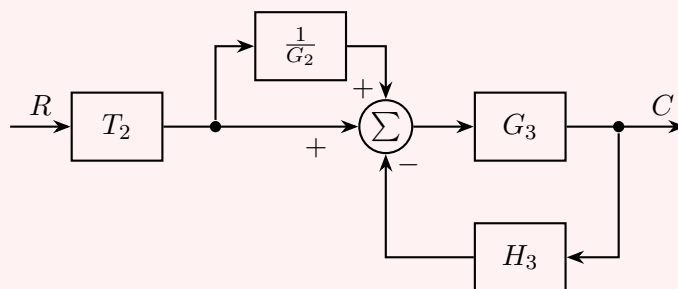
and moving the feedforward branch to after T_1 , we have



With

$$T_2 = \frac{T_1 G_1}{1 + T_1 G_1 H_1}$$

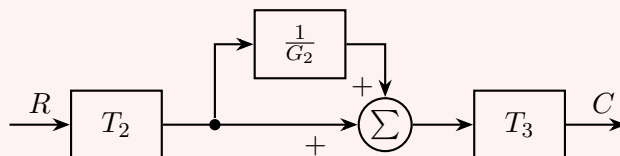
the first section simplifies into a single block:



And finally,

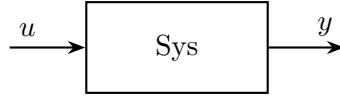
$$T_3 = \frac{G_3}{1 + G_3 H_3}$$

gives a simple parallel-series system to be solved by matlab:



0.6 Sept 29

We can view a system

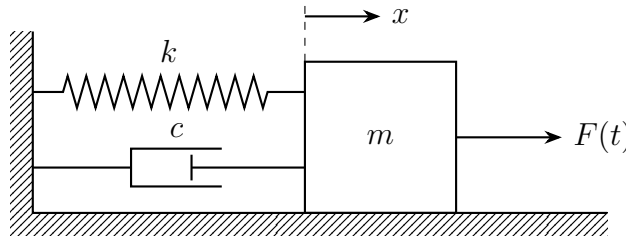


in three ways:

1. Differential Equations (e.g. $\ddot{y} + 2t\dot{y} + y = u$ which cannot be solved via Laplace because it is not LTI)
2. Transfer function (e.g. $\frac{Y}{U} = \frac{s+1}{s^2+4s+11}$)
3. State space (e.g. $\dot{X} = Ax + bu$, $Y = cx + du$)

As it happens, during the Space Race, the Americans were using transfer functions which excelled at single-input, single-output systems. For Sputnik, Lyapunov developed the state-space formulation which can handle *multi-input, multi-output systems*.

Example (converting to state space): Recall the spring-mass-damper system.



$$m\ddot{x} + c\dot{x} + kx = f$$

We want to write this system as a state-space system of the form

$$\begin{cases} \dot{X} = Ax + bu \\ Y = cx + du \end{cases}$$

In general, the order of the differential equation will dictate the size of the matrix A .

For a second-order ODE, $A \in \mathbb{R}^{2 \times 2}$, $u \in \mathbb{R}^{2 \times 1}$, $c \in \mathbb{R}^{1 \times 2}$, $d \in \mathbb{R}^{1 \times 1}$, $x \in \mathbb{R}^{2 \times 1}$.

Define a **state variable**

$$X = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

and suppose $x_1 = x$ (the position of the mass) and $x_2 = \dot{x}$ (the velocity).

In particular, this means

$$\dot{x}_1 = x_2$$

so

$$\ddot{x} = \frac{1}{m}(f - c\dot{x} - kx) \implies \dot{x}_2 = \frac{1}{m}(f - cx_2 - kx_1)$$

hence

$$\begin{aligned} \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} &= \overbrace{\begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{c}{m} \end{bmatrix}}^A \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \overbrace{\begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix}}^B f \\ Y &= \underbrace{\begin{bmatrix} 1 & 0 \end{bmatrix}}_C \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \end{aligned}$$

(making the arbitrary choice that we want the position as our output)

If instead we wanted $[x, \dot{x}]$ as our output, we could define

$$Y = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

This method lets us move from DiffEq to state space. How do we move from state space to transfer function or transfer function to state space?

Trivially, to move from transfer function to state space just cross-multiply to recover the ODE and then move the state space.

The more difficult direction is state space to transfer function. Just take the Laplace and solve for Y/U .

First, solving for $X(s)$:

$$\begin{aligned} sX(s) &= AX(s) + BU(s) \\ (sI - A)X(s) &= BU(s) \\ X(s) &= (sI - A)^{-1}BU(s) \end{aligned}$$

Then, substituting,

$$\begin{aligned} Y(s) &= CX(s) + DU(s) \\ &= (C(sI - A)^{-1}B + D)U(s) \end{aligned}$$

which gives the general form

$$\boxed{\frac{Y(s)}{U(s)} = C(sI - A)^{-1}B + D}$$

In our example,

$$\begin{aligned} (sI - A) &= \begin{bmatrix} s & -1 \\ \frac{k}{m} & s + \frac{c}{m} \end{bmatrix} \\ (sI - A)^{-1} &= \frac{1}{s(s + \frac{c}{m}) + \frac{k}{m}} \begin{bmatrix} s + \frac{c}{m} & 1 \\ -\frac{k}{m} & s \end{bmatrix} \\ \frac{Y(s)}{U(s)} &= C(sI - A)^{-1}B + D \\ &= \frac{1}{s^2 + \frac{cs}{m} + \frac{k}{m}} \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} s + \frac{c}{m} & 1 \\ -\frac{k}{m} & s \end{bmatrix} \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix} \\ &= \frac{1}{s^2 + \frac{cs}{m} + \frac{k}{m}} \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{m} \\ \frac{s}{m} \end{bmatrix} \\ &= \frac{1}{ms^2 + cs + k} \end{aligned}$$

Which is exactly what we found via Laplace:

$$m\ddot{x} + c\dot{x} + kx = f \implies ms^2X(s) + csX(s) + kX(s) = F \implies \frac{X(s)}{F(s)} = \frac{1}{ms^2 + cs + k}$$

The general **non-linear state space form** is

$$\begin{cases} \dot{x} = f(x, u) \\ y = h(x, u) \end{cases}$$

Nonlinear systems can be controlled either via **linearization** or **non-linear control**. Only linearization will be treated in this course.

Two basic methods: make a linear approximation (e.g. $\sin \theta = \theta$) or linearize around equilibria.

Example: Consider a transatlantic flight. Over the length of the flight, fuel usage can lead to a change in mass of almost 40%. We would like our controller to respond to this change without needing to model the system continuously.

We can take advantage of times when the plane has constant speed (e.g. no acceleration) to solve for equilibria

$$\begin{cases} \dot{x} = 0 \\ f(x, u)|_{x_0, u_0} = 0 \\ y = h(x_0, y_0) \end{cases} \implies \begin{matrix} A = \frac{\partial f}{\partial x} \Big|_{u_0, x_0} & C = \frac{\partial h}{\partial x} \Big|_{u_0, x_0} \\ B = \frac{\partial f}{\partial u} \Big|_{u_0, x_0} & D = \frac{\partial h}{\partial u} \Big|_{u_0, x_0} \end{matrix}$$

in a higher dimensional analogue to

$$f(x) = f(x_0) + \frac{\partial f}{\partial x} \Big|_{x_0} (x - x_0)$$